

1. What is Green Chemistry? What are the 12 principles of Green Chemistry?

Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry addresses the environmental impact of both chemical products and the processes by which they are produced. Green chemistry eliminates waste at source, i.e. it is primary pollution prevention rather than waste remediation (end-of-pipe solutions).

The guiding principle is the design of environmentally benign products and processes (benign by design). This concept is embodied in the 12 Principles of Green Chemistry, which are as follows:

- 1. Prevent waste:** Design chemical syntheses to prevent waste. Leave no waste to treat or clean up.
- 2. Maximize atom economy:** Design syntheses so that the final product contains the maximum proportion of the starting materials. Waste few or no atoms.
- 3. Design less hazardous chemical syntheses:** Design syntheses to use and generate substances with little or no toxicity to either humans or the environment.
- 4. Design safer chemicals and products:** Design chemical products that are fully effective yet have little or no toxicity.
- 5. Use safer solvents and reaction conditions:** Avoid using solvents, separation agents, or other auxiliary chemicals. If you must use these chemicals, use safer ones.
- 6. Increase energy efficiency:** Run chemical reactions at room temperature and pressure whenever possible.
- 7. Use renewable feedstocks:** Use starting materials (also known as feedstocks) that are renewable rather than depletable. The source of renewable feedstocks is often agricultural products or the wastes of other processes; the source of depletable feedstocks is often fossil fuels (petroleum, natural gas, or coal) or mining operations.
- 8. Avoid chemical derivatives:** Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste.
- 9. Use catalysts, not stoichiometric reagents:** Minimize waste by using catalytic reactions. Catalysts are effective in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and carry out a reaction only once.
- 10. Design chemicals and products to degrade after use:** Design chemical products to break down to innocuous substances after use so that they do not accumulate in the environment.
- 11. Analyze in real time to prevent pollution:** Include in-process, real-time monitoring and control during syntheses to minimize or eliminate the formation of byproducts.
- 12. Minimize the potential for accidents:** Design chemicals and their physical forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment.

2. Give some examples of use of green chemistry in industrial synthesis.

Now, many industrial applications are found that fulfill several of the 12 principles of green chemistry at the same time.

- a) The most polluting reaction in industry is oxidation. Implementation of green chemistry has led to the use of alternative less polluting reagents viz., metal ion contamination is minimized by using molecular O_2 as the primary oxidant and use of extremely high oxidation state transition metal complexes.
- b) Conventional methylation reactions employing toxic alkyl halides or methylsulfate leading to environmental hazard are replaced by dimethylcarbonate with no deposit of inorganic salts e.g., methylation of arylacetonitriles in presence of K_2CO_3 to 2- arylpropionitriles (> 99%).
- c) The use of phosgene and methylene chloride in the synthesis of polycarbonates has been replaced by diphenylcarbonate.
- d) Use of CO_2 as a reaction medium for asymmetric catalytic reductions particularly hydrogenation and H_2 transfer reactions.
- e) A convenient green synthesis of acetaldehyde is by Wackerchemie's oxidation of ethylene with O_2 in presence of a catalyst, in place of its synthesis by oxidation of ethanol or hydration of acetylene with H_2SO_4 .

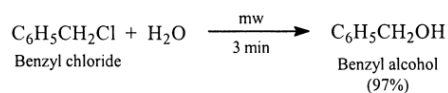
3. Write a note on Green Methodologies such as Microwave and Ultrasound assisted synthesis techniques.

Microwave assisted synthesis

Microwaves have wavelength of 1 mm to 1 m corresponding to frequencies between 0.3 and 300 GHz. Microwave dielectric heating uses the ability of some liquids and solids to transform electromagnetic radiation into heat to drive chemical reactions.

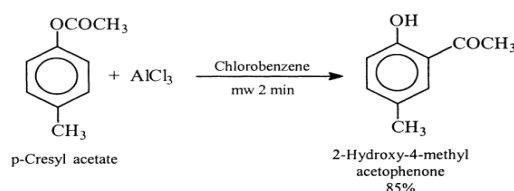
Microwave synthesis is considered as an important approach toward green chemistry, because this technique is more eco-friendly. Due to its ability to couple directly with the reaction molecule and by passing thermal conductivity leading to a rapid rise in the temperature, microwave irradiation has been used to improve many organic syntheses. Microwave chemistry is the science of applying microwave radiation to chemical reactions. Microwave synthesis represents a major breakthrough in synthetic chemistry methodology; a dramatic change in the way chemical synthesis is performed. Conventional heating, long known to be inefficient and time consuming, has been recognized to be creatively limiting too. Instead of spending hours or even days synthesizing a single compound, chemists can now perform the same reaction in minutes. The problem associated with waste disposal of solvents has been overcome by performing reactions without a solvent under microwave irradiation. Coupling of microwave irradiation with the use of mineral-supported catalysed reactions, under solvent-free conditions, provides clean chemical processes with the advantage of enhanced reaction rates, higher yields, greater selectivity, and greater ease of manipulation. Thus, microwave synthesis acts as a potential tool for green chemistry. Microwave irradiation provides an alternative to the conventional methods, for heating or introducing energy into the system. It utilizes the ability of mobile electric charges present in liquid or conducting ions in solid to transform electromagnetic energy into heat. Microwave radiations are electromagnetic waves. In the electromagnetic spectrum, the microwave radiation region is located between infrared radiation and radio waves.

Examples: Hydrolysis of Benzyl Chloride Hydrolysis of benzyl chloride with water in microwave oven gives 97% yield of benzyl alcohol in 3 min (Scheme 2). The usual hydrolysis in normal way takes about 35 min.



Scheme 2

Fries Rearrangement Fries rearrangement is a useful method for the preparation of phenolic ketones and is usually carried out by heating a mixture of substrate and aluminium chloride. There is considerable rate enhancement of Fries rearrangement by commercial microwave ovens over conventional methods. Thus, a mixture of p-cresyl acetate and anhydrous aluminium chloride are heated in dry chlorobenzene in a sealed tube in a microwave oven for 2 min to give 85% yield of the product (Scheme 10).

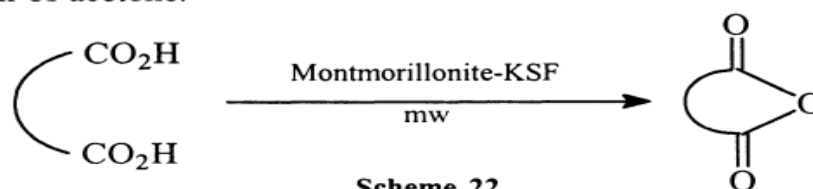


Scheme 10

Example of solvent less mw reaction

9.2.3.7 Synthesis of Anhydrides from Dicarboxylic Acid

Dicarboxylic acids can be converted into anhydrides (Scheme 22) in the presence of isopropenyl acid (which acts as a water scavenger) under microwave irradiation³⁴ using montmorillonite-KSF. The driving force is the formation of acetone.



Scheme 22

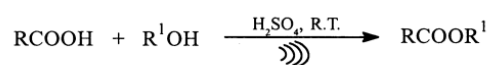
Ultrasound assisted synthesis

The ultrasound frequencies of interest for chemical reactions are in the range of about 20-100 KHz. The term 'sonochemistry' is used to describe the effect of ultrasound waves on chemical reactivity.

When a sound wave, propagated by a series of compression and refraction cycles, pass through a liquid medium, it causes the molecules to oscillate around their mean position. During the compression cycle, the average distance between the molecules is reduced and during refraction, the average distance between the molecules is increased. In the refraction cycle, under appropriate conditions, the attractive forces of the molecules of the liquid may be overcome, causing formation of bubbles. In case the internal forces are great enough to ensure collapse of these bubbles, very high local temperature (around

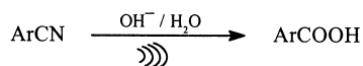
5000°C) and pressure (over 1000 bar) may be created. It is this very high temperature' and pressure that initiate chemical reactions.

Example: Esterification This is generally carried out in presence of a catalyst like sulphuric acid, p-toluenesulphonic acid, tosyl chloride, polyphosphoric acid, dicyclohexylcarbodiimide etc. The reaction takes longer time and yields are low. A simple procedure for the esterification of a variety of carboxylic acids with different alcohols at ambient temperature using ultrasound has been reported (Scheme 1).



Scheme 1

Hydrolysis Nitriles can be hydrolysed to carboxylic acids (Scheme 3) under basic condition on sonication.



Scheme 3

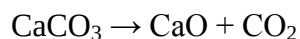
4. Describe the atom economy concept in detail. What is the impact of atom economy on the environment?

Atom economy means maximizing the incorporation of material from the starting materials or reagents into the final product. It is essentially pollution prevention at the molecular level. **Atom economy is defined as:**

$$\% \text{Atom economy} = (\text{Mass of wanted product(s)} / \text{Total mass of products}) \times 100$$

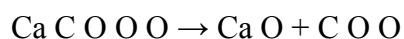
The idea of yield of a product is useful, but from a Green Chemistry and sustainable development perspective, it is not the full picture. This is because yield is calculated by considering only one reactant and one product. One of the key principles of Green Chemistry is that processes should be designed so that the maximum amount of all the raw materials ends up in the product and a minimum amount of waste is produced. A reaction can have a high percentage yield but also make a lot of waste product. This kind of reaction has a low atom economy. Both the yield and the atom economy have to be taken into account when designing a green chemical process.

For example:



RMM or Mr: 100 56 44

If we split up the formulae, we can look at what happens to each atom in the reaction. The atoms shown below in bold end up in the product we want, the rest do not:



Waste box: 1C 2O

From the original atoms, one C atom and two O atoms are wasted – they are not in the final, useful product.

So for this example, % Atom economy = $(56 \times 100)/100 = 56\%$.

Cleaning up the environment and, more importantly, preventing pollution are important issues in today's world. While the chemical industry is traditionally viewed more as a cause than a solution to pollution, chemistry does offer unique solutions in the area of waste prevention. One of the most fundamental of these solutions is the application of the green chemistry principle of atom economy to chemical reactions. Atom economy moves the practice of minimizing waste to the molecular level. Traditionally, chemists have focused on maximizing yield, minimizing the number of steps or synthesizing a completely unique chemical. Green chemistry and atom economy introduce a new goal into reaction chemistry: designing reactions so that the atoms present in the starting materials end up in the product rather than in the wastestream.

5. What are some of the emerging/new trends in green chemistry?

1. Oxidative chemical agents and catalysts Several of the oxidizing reagents and catalysts are comprised of nephrotoxic (damaging or destructive to the kidneys) substances like metals. Since these substances are typically employed in very massive volumes, a huge amount of these metals have been discharged in the environment and have substantial negative impact on the health of all living beings. These may be substituted by more eco-friendly substances.

2. Computational chemistry is a branch of chemistry that uses computer simulation to assist in solving chemical problems. It uses methods of theoretical chemistry, incorporated into efficient computer programs, to calculate the structures and properties of molecules and solids. For e.g., various different derivatives of the toxic element lead (Pb) can be virtually created and their different properties can be assessed/predicted, without the consequences of hazardous waste creation and disposal.

3. Proliferation of solventless reactions The procedure of solvent-less condition utilizes some non-traditional conditions, such as microwave heating or ultrasonication techniques. This helps in development of product isolation, separation, and purification that may be solventless.

4. Use of green solvents and environmental conditions for carrying out reactions Current trends in synthetic chemistry focus towards the development of reaction routes that make use of eco-friendly solvents such as water and ethanol. Reactions that can take place at room temperature conditions or ambient conditions such as in the presence of sunlight or UV rays (photochemical reactions) are much preferred over those requiring laboratory set ups.

6. Define Industrial Ecology and list its applications.

Industrial ecology (IE) is a branch of systems science for sustainability, or a framework for designing and operating industrial systems as sustainable and interdependent with natural systems. It seeks to balance industrial production and economic performance with an emerging understanding of local and global ecological constraints.

IE provides a foundation for sustainable industrialization, not just incremental improvement in environmental management. The objectives of IE suggest a potential for reindustrialization in economies that have lost major components of their industrial base. Specifically, the objective of industrial ecology is not merely to reduce pollution and waste as traditionally conceived, it is to reduce throughput of all kinds of materials and fuels, whether they leave a site as products, emissions, or waste.

Industrial ecology is applied to the management of human activity on a sustainable basis by:

- (a) minimizing energy and materials usage;
- (b) ensuring acceptable quality of life for people;
- (c) minimizing the ecological impact of human activity to levels natural systems can sustain;
- (d) conserving and restoring ecosystem health and maintaining biodiversity;
- (e) maintaining the economic viability of systems for industry, trade, and commerce;
- (f) coordinating design over the life cycle of products and processes; and
- (g) enabling creation of short-term innovations with awareness of their long-term impacts.

Application of IE will improve the planning and performance of industrial systems of all sizes, and will help design local and community solutions that contribute to national and global solutions.

7. What are the different kinds of soil microorganisms? What are their functions and importance?

Soil microorganisms can be grouped on the basis of:

–Size

–Species

–Function

Soil Bacteria There are many types of bacteria but the focus here is on those that are important for agriculture, e.g. Rhizobium and actinomycetes. Bacteria are important in agricultural soils because they contribute to the carbon cycle by fixation (photosynthesis) and decomposition. Some bacteria are important decomposers and others such as actinomycetes are particularly effective at breaking down tough substances such as cellulose (which makes up the cell walls of plants) and chitin (which makes up the cell walls of fungi).

Soil Actinomycetes

1. Transitional group between bacteria and fungi
2. Active in degrading more resistant organic compounds

3. Optimal growth at alkaline pH

4. 2 important products: – produce antibiotics (streptomycin is produced by an actino) and produce geosmin (an organic compound with a distinct earthy flavor and aroma produced by certain bacteria, and is responsible for the earthy taste of beetroots and a contributor to the strong scent (petrichor) that occurs in the air when rain falls after a dry spell of weather or when soil is disturbed).

5. Negative impact - potato scab (*Streptomyces scabies*)

Soil Fungi These organisms are responsible for the important process of decomposition in terrestrial ecosystems as they degrade and assimilate cellulose, the component of plant cell walls. Fungi are constituted by microscopic cells that usually grow as long threads or strands called hyphae of only a few micrometres in diameter but with the ability to span a length from a few cells to many metres.

1. Dominate the soil biomass

2. Obligate aerobes

3. Can survive desiccation

4. Dominate in acid soils

5. Negative impacts: – Apple replant disease (*Rhizoctonia*, *Pythium*, *Fusarium*, and *Phytophthora*) – Powdery mildew is caused by a fungus

6. Beneficials:– *Penicillium*

Soil Protozoa These are organisms that have only one cell, and are microscopic, but larger than bacteria. They are grouped by the ways they move: amoebae use a psuedo (fake) foot, ciliates have cilia (short hair) and move them very fast, and flagellates have one or more flagella (whips) and move them very fast.

Functions: As they eat bacteria, protozoa release excess nitrogen that can then be used by plants and other members of the food web. Protozoa play an important role in mineralizing nutrients, making them available for use by plants and other soil organisms. Protozoa have a lower concentration of nitrogen in their cells than the bacteria they eat. (The ratio of carbon to nitrogen for protozoa is 10:1 or much more and 3:1 to 10:1 for bacteria.) Bacteria eaten by protozoa contain too much nitrogen for the amount of carbon protozoa need. They release the excess nitrogen in the form of ammonium (NH_4^+). This usually occurs near the root system of a plant. Bacteria and other organisms rapidly take up most of the ammonium, but some is used by the plant. Another role that protozoa play is in regulating bacteria populations. When they graze on bacteria, protozoa stimulate growth of the bacterial population (and, in turn, decomposition rates and soil aggregation.) Protozoa are also an important food source for other soil organisms and help to suppress disease by competing with or feeding on pathogens

Soil Nematodes (Worms, typically 1/500 of an inch (50 µm) in diameter and 1/20 of an inch (1 mm) in length.) An incredible variety of nematodes function at several trophic levels of the soil food web. Some feed on the plants and algae (first trophic level); others are grazers that feed on bacteria and fungi (second trophic level); and some feed on other nematodes (higher trophic levels). Free-living nematodes can be divided into four broad groups based on their diet. Bacterial-feeders consume bacteria. Fungal-feeders feed by puncturing the cell wall of fungi and sucking out the internal contents. Predatory nematodes eat all types of nematodes and protozoa. They eat smaller organisms whole, or attach themselves to the cuticle of larger nematodes, scraping away until the prey's internal body parts can be extracted.

Functions: Nutrient cycling. Like protozoa, nematodes are important in mineralizing, or releasing, nutrients in plant-available forms. When nematodes eat bacteria or fungi, ammonium (NH_4^+) is released because bacteria and fungi contain much more nitrogen than the nematodes require.

Importance of Soil Microorganisms

- Responsible for cycling of C, N, and other nutrients
- Enhance soil structure
- Relocate and decompose organic materials
- Maintain soil quality and health
- Increase soil aeration and penetrability
- Involved in disease transmission and control

8. Differentiate between aerobic and anaerobic composting giving examples of the microorganisms involved in each case.